

LECTURE 08

DYNAMIC FUNCTIONS IN EXCEL



POWERFUL FORMULAS.
DYNAMIC RESULTS.
SMARTER ANALYSIS.

EXCEL 365 • SMARTER • FASTER • DYNAMIC

ONE FORMULA.
MULTIPLE RESULTS.
AUTOMATICALLY.



DYNAMIC ARRAYS

Return multiple values and automatically spill the results.



KEY DYNAMIC FUNCTIONS



FILTER()

Extracts rows that meet your criteria.



SORT()

Sorts data in ascending or descending order.



SORTBY()

Sorts one range based on another range.



UNIQUE()

Returns unique values automatically.



SEQUENCE()

Generates sequential numbers.



RANDARRAY()

Creates random numbers in an array.

WHY USE DYNAMIC FUNCTIONS?



SAVE TIME

One formula does the work of many.



DYNAMIC RESULTS

Results update automatically with data.



ACCURATE ANALYSIS

Reduce errors and improve data reliability.



POWERFUL REPORTS

Build interactive dashboards with ease.



INTERVIEW READY

Essential skills for modern data analysts.



DYNAMIC THINKING.
DYNAMIC FORMULAS.
DYNAMIC RESULTS.

Lecture 08 – Dynamic Array Functions in Excel

Module Objective

- Understand Dynamic Arrays.
- Learn how Excel spills formulas automatically.
- Master modern Excel functions used by Data Analysts.
- Build dynamic reports with fewer formulas.
- Replace traditional formulas with Dynamic Array functions.

Module 1 – Introduction to Dynamic Arrays

What are Dynamic Array Functions?

Dynamic Array Functions are modern Excel functions that can return **multiple values** from a single formula. Instead of returning one value, Excel automatically spills the result into adjacent cells.

Definition

A Dynamic Array Function returns one or more values from a single formula, and Excel automatically fills the required cells.

Traditional Excel

Before Excel 365

One Formula

↓

One Result

Example

=A2

returns

Laptop

Dynamic Excel

One Formula

↓

Multiple Results

Example

=FILTER(A2:D20,D2:D20="North")

returns

All North Region records.

What is Spill?

When Excel automatically expands a formula into multiple cells, it is called **Spilling**.



Example

Formula

=SEQUENCE(10)

Result

1
2
3
4
5
6
7
8
9
10

Only one formula was entered.

Excel filled the remaining cells automatically.

Spill Range

The cells occupied by a Dynamic Array formula are known as the **Spill Range**.

Spill Operator (#)

Excel provides a Spill Reference.

Example

=A2#

It refers to the entire spilled range.

Why Are Dynamic Arrays Important?

Traditional Excel required:

- Copy formulas
- Drag formulas
- Fill down

Dynamic Arrays perform everything automatically.

Business Benefits

- Less manual work
- Faster reports
- Smaller formulas
- Dynamic dashboards
- Better automation
- Easier maintenance

Module 2 – Dynamic Array Functions Overview

Excel provides several Dynamic Array functions.



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We'll cover:

- FILTER
- SORT
- SORTBY
- UNIQUE
- SEQUENCE
- RANDARRAY

These functions are heavily used in Data Analytics and dashboard reporting.

Module 3 – FILTER Function

Definition

Returns only the rows that satisfy a specified condition.

Purpose

Extract records matching one or more business conditions.

Syntax

`=FILTER(array, include, [if_empty])`

Arguments

array

The data range to filter.

include

Logical condition.

Must return TRUE or FALSE.

if_empty

Optional.

Message displayed when no records match.

Business Example

Dataset

Region Revenue

North 125000

South 54000

North 89000

West 140000

Formula

`=FILTER(A2:B20,A2:A20="North")`

Result

Returns only North region records.



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Business Uses

- Customer filtering
- Product filtering
- Employee filtering
- Region-wise analysis
- Sales reports
- Dashboard reports

Interview Question

Why use FILTER instead of Auto Filter?

Answer:

FILTER is dynamic, formula-based, automatically updates, and can feed dashboards, whereas Auto Filter is manual and primarily for viewing data.

Module 4 – SORT Function

Definition

Sorts data dynamically without changing the original dataset.

Purpose

Create sorted reports while keeping source data intact.

Syntax

`=SORT(array,[sort_index],[sort_order],[by_col])`

Arguments

array

Dataset.

sort_index

Column number to sort.

sort_order

1 → Ascending

-1 → Descending

by_col

FALSE

(Default)

Sort by rows.

TRUE

Sort by columns.

Business Example



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Sort Revenue

Highest to Lowest

Formula

`=SORT(A2:E20,5,-1)`

Business Uses

- Sales Ranking
- Customer Ranking
- Product Ranking
- Dashboard Reports

Why is SORT useful?

Traditional sorting changes the original data.

SORT creates a new dynamic sorted report.

Module 5 – SORTBY Function

Definition

Sorts one dataset based on another column or range.

Unlike SORT, you don't need to specify the column number.

Syntax

`=SORTBY(array,by_array,[sort_order])`

Arguments

array

Data to return.

by_array

Column used for sorting.

sort_order

1 → Ascending

-1 → Descending

Business Example

Sort employees based on Revenue.

`=SORTBY(A2:D20,D2:D20,-1)`

Result

Employees arranged by highest Revenue.

Why SORTBY?

If columns move, SORT may break because it relies on column numbers.

SORTBY uses ranges, making it more flexible.



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Business Uses

- Employee ranking
- Product ranking
- KPI dashboards
- Monthly reports

Comparison

Function Best Use

FILTER Extract records

SORT Sort using column index

SORTBY Sort using another range

Best Practices

- ✓ Use Excel Tables with Dynamic Arrays.
- ✓ Use Spill References (#) wherever possible.
- ✓ Avoid typing inside spill ranges.
- ✓ Combine Dynamic Array functions with XLOOKUP and IFERROR for powerful reports.
- ✓ Prefer SORTBY over SORT when datasets may change structure.



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Part 02 – UNIQUE, SEQUENCE, RANDARRAY & Advanced Dynamic Arrays

Module 6 – UNIQUE Function

What is the UNIQUE Function?

The **UNIQUE** function returns only the distinct (non-duplicate) values from a dataset. Instead of manually removing duplicates or using the Remove Duplicates feature, Excel automatically generates a dynamic list of unique values.

Definition

Returns unique values or records from a specified range.

Purpose

To create dynamic lists without duplicate entries.

Syntax

=UNIQUE(array,[by_col],[exactly_once])

Arguments

array

The range or table containing the data.

by_col (Optional)

FALSE (Default)

Checks uniqueness row-wise.

TRUE

Checks uniqueness column-wise.

exactly_once (Optional)

FALSE

Returns all distinct values.

TRUE

Returns only values that appear exactly once.

Business Example

Region Column

Region

North

South

North

East

West

South



Formula

=UNIQUE(A2:A7)

Result

North

South

East

West

Business Use Cases

- Customer List
- Employee List
- Product Categories
- Cities
- Regions
- Departments

Interview Question

Why use UNIQUE instead of Remove Duplicates?

Answer

Remove Duplicates permanently changes the source data.

UNIQUE creates a **dynamic list** that updates automatically whenever new data is added.

Module 7 – SEQUENCE Function

What is SEQUENCE?

Creates a sequence of numbers automatically.

Definition

Returns a list of sequential numbers.

Syntax

=SEQUENCE(rows,[columns],[start],[step])

Arguments

rows

Number of rows.

columns (Optional)

Number of columns.

Default = 1

start (Optional)



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Starting number.

Default = 1

step (Optional)

Increment value.

Default = 1

Example 1

=SEQUENCE(10)

Result

1
2
3
4
5
6
7
8
9
10

Example 2

=SEQUENCE(5,1,100,10)

Result

100
110
120
130
140

Business Uses

- Invoice Numbers
 - Employee IDs
 - Serial Numbers
 - Dashboard Numbering
 - Report Indexing
-

Business Scenario

Generate Invoice IDs automatically

INV-1001

INV-1002

INV-1003



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Module 8 – RANDARRAY Function

Definition

Returns an array of random numbers.

Purpose

Generate sample datasets.

Testing.

Simulation.

Practice.

Syntax

=RANDARRAY([rows],[columns],[min],[max],[whole_number])

Arguments

rows

Number of rows.

columns

Number of columns.

min

Minimum value.

max

Maximum value.

whole_number

TRUE

Whole numbers

FALSE

Decimals

Example

=RANDARRAY(10,1,1000,5000,TRUE)

Result

Ten random numbers

Between

1000

and

5000

Business Uses

- Sample Data
- Testing Dashboards
- Sales Simulation



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- Inventory Simulation
- Forecast Practice

Module 9 – Combining Dynamic Functions

Dynamic Arrays become powerful when combined.

FILTER + SORT

Business Requirement

Show North Region orders

sorted by Revenue

Highest to Lowest.

Formula

`=SORT(FILTER(A2:H100,C2:C100="North"),8,-1)`

Business Benefit

One formula

↓

Dynamic report

FILTER + UNIQUE

Requirement

Show all unique products sold in North Region.

Formula

`=UNIQUE(FILTER(E2:E100,C2:C100="North"))`

Business Uses

Regional Product Analysis

Inventory Planning

Sales Planning

SORT + UNIQUE

Requirement

Display customer names alphabetically without duplicates.

Formula

`=SORT(UNIQUE(B2:B500))`

Business Uses

CRM Reports

Customer Lists

Email Campaigns

FILTER + SORT + UNIQUE

Formula



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=SORT(UNIQUE(FILTER(E2:E500,C2:C500="West")))

Result

Unique products



West Region



Alphabetically Sorted

Business Uses

Regional Inventory Reports

Regional Sales Reports

Warehouse Reports

Module 10 – Dynamic Dashboard Applications

Modern dashboards heavily rely on Dynamic Arrays.

Business Dashboard

User selects

North



FILTER



Displays only North records.

User changes

South



Dashboard updates automatically.

No manual filtering required.

Dashboard Uses

Sales Dashboard

HR Dashboard

Finance Dashboard

Marketing Dashboard

Inventory Dashboard

Executive Dashboard

Module 11 – Common Spill Errors

Sometimes Dynamic Arrays cannot expand.



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Excel displays
#SPILL!

Causes

1

Cells already contain data.

2

Merged Cells

3

Excel Tables restricting spill area.

4

Insufficient space.

Solution

Clear surrounding cells.

Unmerge cells.

Increase available space.

Module 12 – Dynamic Arrays vs Traditional Excel

Traditional Excel

Dynamic Arrays

Copy formulas manually One formula spills automatically

Static reports

Dynamic reports

Manual updates

Automatic updates

Complex formulas

Simpler formulas

More maintenance

Easier maintenance

Real Business Case Study

Scenario

A retail company receives **25,000 transactions every month**.

Management requires:

- Region-wise reports
- Unique customer lists
- Top-selling products
- Highest revenue orders
- Dynamic monthly reports

Using Dynamic Array Functions:

- FILTER() generates region-specific reports.
- UNIQUE() creates customer and product lists.
- SORTBY() ranks salespersons by revenue.
- SEQUENCE() generates report serial numbers.



- RANDARRAY() creates sample datasets for testing.

The reports update automatically as new records are added, eliminating manual filtering and sorting.

Best Practices

- ✓ Use Excel Tables for expanding datasets.
 - ✓ Prefer Dynamic Arrays over copied formulas.
 - ✓ Combine with XLOOKUP, IFERROR, and Conditional Formatting.
 - ✓ Reference spilled ranges using #.
 - ✓ Keep spill ranges free of data.
 - ✓ Use descriptive headers for dynamic reports.
-

Interview-Oriented Questions (with Answers)

Q1. What are Dynamic Array Functions?

Answer: Functions that return multiple values from a single formula, automatically spilling results into adjacent cells.

Q2. What is a Spill Range?

Answer: The group of cells automatically filled by a Dynamic Array formula.

Q3. What does the # operator represent?

Answer: It references the entire spilled range of a Dynamic Array formula.

Example:

=A2#

Q4. Difference between SORT() and SORTBY()?

Answer: SORT() sorts using a column index, whereas SORTBY() sorts using another range, making it more flexible when columns are rearranged.

Q5. Why is UNIQUE() better than Remove Duplicates?

Answer: UNIQUE() creates a dynamic list that updates automatically without modifying the original dataset.

Q6. Give one business use case for FILTER().

Answer: Display all transactions for the "North" region in a dashboard that updates automatically when the source data changes.

Q7. What causes a #SPILL! error?

Answer: Existing data blocking the spill range, merged cells, or insufficient space for the array to expand.

Q8. How can SEQUENCE() be used in business reporting?

Answer: To generate serial numbers, invoice numbers, report indexes, or row numbering dynamically.

Q9. What is the purpose of RANDARRAY()?

Answer: It generates random values for simulations, testing dashboards, sample datasets, and training exercises.

Q10. How do Dynamic Arrays improve dashboard development?

Answer: They eliminate manual copying of formulas, automatically update reports when data changes, simplify formulas, and make dashboards interactive and easier to maintain.



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